

MongoDB Complete Commands – Student Collection

```
>_MONGOSH

> show dbs
< admin          40.00 KiB
  bookflow_db    100.00 KiB
  config         72.00 KiB
  local          72.00 KiB
> use studentDB
< switched to db studentDB
> db
< studentDB
> show collections
<
> db.createCollection("students")
< { ok: 1 }
> db.students.insertMany([
  { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
  { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
  { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
  { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
  { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
  { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
  { rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
  { rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
  { rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
  { rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
])
```

>_MONGOSH

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a94ec4fdce903635e4dd5fc'),
    '1': ObjectId('6a94ec4fdce903635e4dd5fd'),
    '2': ObjectId('6a94ec4fdce903635e4dd5fe'),
    '3': ObjectId('6a94ec4fdce903635e4dd5ff'),
    '4': ObjectId('6a94ec4fdce903635e4dd600'),
    '5': ObjectId('6a94ec4fdce903635e4dd601'),
    '6': ObjectId('6a94ec4fdce903635e4dd602'),
    '7': ObjectId('6a94ec4fdce903635e4dd603'),
    '8': ObjectId('6a94ec4fdce903635e4dd604'),
    '9': ObjectId('6a94ec4fdce903635e4dd605')
  }
}
> db.students.insertOne({
  rollNo: 111,
  name: "Vijay",
  age: 21,
  department: "CSE",
  marks: 86,
  city: "Hyderabad"
})
```

```
>_MONGOSH
< {
  acknowledged: true,
  insertedId: ObjectId('6a94eca9dce903635e4dd606')
}
> db.students.insertMany([
  { rollNo: 112, name: "Pooja", age: 20, department: "IT", marks: 84 },
  { rollNo: 113, name: "Ramesh", age: 22, department: "ECE", marks: 79 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a94ecb4dce903635e4dd607'),
    '1': ObjectId('6a94ecb4dce903635e4dd608')
  }
}
> db.students.find()
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
```

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```
{
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5fe'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
```

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```
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd600'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd601'),  
  rollNo: 106,  
  name: 'Anu',  
  age: 20,  
  department: 'IT',  
  marks: 82,  
  city: 'Guntur'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd602'),  
  rollNo: 107,  
  name: 'Ravi',  
  age: 23,  
  department: 'ECE',  
  marks: 68,
```

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```
    city: 'Nellore'
  }
  {
    _id: ObjectId('6a94ec4fdce903635e4dd603'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94ec4fdce903635e4dd604'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  }
  {
    _id: ObjectId('6a94ec4fdce903635e4dd605'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
```

```
>_MONGOSH
marks: 93,
city: 'Hyderabad'
}
{
  _id: ObjectId('6a94eca9dce903635e4dd606'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd607'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd608'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
```

```
>_MONGOSH
```

```
marks: 79
```

```
}
```

```
> db.students.findOne()
```

```
< {
```

```
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
```

```
  rollNo: 101,
```

```
  name: 'John',
```

```
  age: 20,
```

```
  department: 'CSE',
```

```
  marks: 85,
```

```
  city: 'Hyderabad'
```

```
}
```

```
> db.students.findOne({ rollNo: 101 })
```

```
< {
```

```
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
```

```
  rollNo: 101,
```

```
  name: 'John',
```

```
  age: 20,
```

```
  department: 'CSE',
```

```
  marks: 85,
```

```
  city: 'Hyderabad'
```

```
}
```

```
>_MONGOSH
```

```
> db.students.find({ department: "CSE" })
```

```
< {
```

```
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
```

```
  rollNo: 101,
```

```
  name: 'John',
```

```
  age: 20,
```

```
  department: 'CSE',
```

```
  marks: 85,
```

```
  city: 'Hyderabad'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),
```

```
  rollNo: 102,
```

```
  name: 'Alice',
```

```
  age: 21,
```

```
  department: 'CSE',
```

```
  marks: 90,
```

```
  city: 'Vijayawada'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a94ec4fdce903635e4dd600'),
```

```
  rollNo: 105,
```

```
  name: 'Kiran',
```

```
  age: 21,
```

```
  department: 'CSE',
```

```
  marks: 95,
```

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```
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd603'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94eca9dce903635e4dd606'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
```

>_MONGOSH

```
> db.students.find({ name: "John" })
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
> db.students.countDocuments()
< 13
> db.students.countDocuments({ department: "CSE" })
< 5
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
```

>_MONGOSH

```
{
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd600'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
```

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```
}  
  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd601'),  
  rollNo: 106,  
  name: 'Anu',  
  age: 20,  
  department: 'IT',  
  marks: 82,  
  city: 'Guntur'  
}  
  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd603'),  
  rollNo: 108,  
  name: 'Sita',  
  age: 21,  
  department: 'CSE',  
  marks: 91,  
  city: 'Hyderabad'  
}  
  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd604'),  
  rollNo: 109,  
  name: 'Arun',  
  age: 22,  
  department: 'AI&DS',  
  marks: 87,  
  city: 'Hyderabad'
```

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```
  city: 'Kurnool'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd605'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94eca9dce903635e4dd606'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd607'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
```

>_MONGOSH

```
marks: 84
}
> db.students.find({ marks: { $gte: 80 } })
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
```

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```
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  }
  {
    _id: ObjectId('6a94ec4fdce903635e4dd600'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94ec4fdce903635e4dd601'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  }
  {
    _id: ObjectId('6a94ec4fdce903635e4dd603'),
    rollNo: 108,
    name: 'Sita',
```

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```
    age: 21,  
    department: 'CSE',  
    marks: 91,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a94ec4fdce903635e4dd604'),  
    rollNo: 109,  
    name: 'Arun',  
    age: 22,  
    department: 'AI&DS',  
    marks: 87,  
    city: 'Kurnool'  
  }  
  {  
    _id: ObjectId('6a94ec4fdce903635e4dd605'),  
    rollNo: 110,  
    name: 'Meena',  
    age: 20,  
    department: 'AI&DS',  
    marks: 93,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a94eca9dce903635e4dd606'),  
    rollNo: 111,
```

```
>_MONGOSH
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd607'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
> db.students.find({ marks: { $lt: 80 } })
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fe'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd602'),
  rollNo: 107
```

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```
  rollNo: 107,  
  name: 'Ravi',  
  age: 23,  
  department: 'ECE',  
  marks: 68,  
  city: 'Nellore'  
}  
{  
  _id: ObjectId('6a94ecb4dce903635e4dd608'),  
  rollNo: 113,  
  name: 'Ramesh',  
  age: 22,  
  department: 'ECE',  
  marks: 79  
}
```

> db.students.find({ marks: { \$lte: 80 } })

```
< {  
  _id: ObjectId('6a94ec4fdce903635e4dd5fe'),  
  rollNo: 103,  
  name: 'Rahul',  
  age: 20,  
  department: 'ECE',  
  marks: 75,  
  city: 'Hyderabad'  
}  
{
```

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```
  _id: ObjectId('6a94ec4fdce903635e4dd602'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd608'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
> db.students.find({ department: { $ne: "CSE" } })
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fe'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
```

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```
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd601'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd602'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
```

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```
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd604'),  
  rollNo: 109,  
  name: 'Arun',  
  age: 22,  
  department: 'AI&DS',  
  marks: 87,  
  city: 'Kurnool'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd605'),  
  rollNo: 110,  
  name: 'Meena',  
  age: 20,  
  department: 'AI&DS',  
  marks: 93,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94ecb4dce903635e4dd607'),  
  rollNo: 112,  
  name: 'Pooja',  
  age: 20,  
  department: 'IT',  
  marks: 84
```

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```
}  
{  
  _id: ObjectId('6a94ecb4dce903635e4dd608'),  
  rollNo: 113,  
  name: 'Ramesh',  
  age: 22,  
  department: 'ECE',  
  marks: 79  
}  
> db.students.find({ marks: { $eq: 90 } })  
< {  
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),  
  rollNo: 102,  
  name: 'Alice',  
  age: 21,  
  department: 'CSE',  
  marks: 90,  
  city: 'Vijayawada'  
}  
> db.students.find({  
  $and: [  
    { department: "CSE" },  
    { marks: { $gt: 85 } }  
  ]  
})  
< {
```

>_MONGOSH

```
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),  
  rollNo: 102,  
  name: 'Alice',  
  age: 21,  
  department: 'CSE',  
  marks: 90,  
  city: 'Vijayawada'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd600'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd603'),  
  rollNo: 108,  
  name: 'Sita',  
  age: 21,  
  department: 'CSE',  
  marks: 91,  
  city: 'Hyderabad'  
}
```

>_MONGOSH

```
{  
  _id: ObjectId('6a94eca9dce903635e4dd606'),  
  rollNo: 111,  
  name: 'Vijay',  
  age: 21,  
  department: 'CSE',  
  marks: 86,  
  city: 'Hyderabad'  
}
```

```
> db.students.find(  
  $or: [  
    { department: "CSE" },  
    { department: "IT" }  
  ]  
)
```

```
< {  
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),  
  rollNo: 101,  
  name: 'John',  
  age: 20,  
  department: 'CSE',  
  marks: 85,  
  city: 'Hyderabad'  
}  
  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),  
  rollNo: 102
```

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```
  rollNo: 102,  
  name: 'Alice',  
  age: 21,  
  department: 'CSE',  
  marks: 90,  
  city: 'Vijayawada'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),  
  rollNo: 104,  
  name: 'Priya',  
  age: 22,  
  department: 'IT',  
  marks: 88,  
  city: 'Warangal'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd600'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{
```

>_MONGOSH

```
{
  _id: ObjectId('6a94ec4fdce903635e4dd601'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd603'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94eca9dce903635e4dd606'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
```

```
>_MONGOSH
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd607'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
> db.students.find({
  marks: { $not: { $gt: 80 } }
})
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fe'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd602'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
```

```
>_MONGOSH

  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd608'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
> db.students.find({
  $nor: [
    { department: "CSE" },
    { marks: { $gt: 90 } }
  ]
})
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fe'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
```

>_MONGOSH

```
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd601'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd602'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
```

>_MONGOSH

```
{
  _id: ObjectId('6a94ec4fdce903635e4dd604'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd607'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a94ecb4dce903635e4dd608'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
```

```
>_MONGOSH
> db.students.find({
  department: { $in: ["CSE", "IT"] }
})
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT'.
```

>_MONGOSH

```
marks: 88,  
city: 'Warangal'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd600'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd601'),  
  rollNo: 106,  
  name: 'Anu',  
  age: 20,  
  department: 'IT',  
  marks: 82,  
  city: 'Guntur'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd603'),  
  rollNo: 108,  
  name: 'Sita',  
  age: 21,
```

>_MONGOSH

```
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94eca9dce903635e4dd606'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94ecb4dce903635e4dd607'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  }
> db.students.find({
  department: { $in: ["CSE", "IT"] }
})
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  --
```

>_MONGOSH

```
< {
  _id: ObjectId('6a94ec4fdce903635e4dd5fc'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5fd'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94ec4fdce903635e4dd5ff'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
```

>_MONGOSH

```
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd600'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd601'),  
  rollNo: 106,  
  name: 'Anu',  
  age: 20,  
  department: 'IT',  
  marks: 82,  
  city: 'Guntur'  
}  
{  
  _id: ObjectId('6a94ec4fdce903635e4dd603'),  
  rollNo: 108,  
  name: 'Sita',  
  age: 21,  
  department: 'CSE',  
  marks: 91,
```

>_MONGOSH

```
marks: 92,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94eca9dce903635e4dd606'),  
  rollNo: 111,  
  name: 'Vijay',  
  age: 21,  
  department: 'CSE',  
  marks: 86,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94ecb4dce903635e4dd607'),  
  rollNo: 112,  
  name: 'Pooja',  
  age: 20,  
  department: 'IT',  
  marks: 84  
}  
> db.students.updateOne(  
  { rollNo: 101 },  
  { $set: { marks: 92 } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,
```

>_MONGOSH

```
    matchedCount: 1,  
    modifiedCount: 1,  
    upsertedCount: 0  
  }  
> db.students.updateMany(  
  { department: "CSE" },  
  { $set: { status: "Active" } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 5,  
  modifiedCount: 5,  
  upsertedCount: 0  
}  
> db.students.updateOne(  
  { rollNo: 101 },  
  { $inc: { marks: 5 } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

>_MONGOSH

```
> db.students.updateOne(
  { rollNo: 101 },
  { $inc: { marks: -2 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.updateMany(
  {},
  { $rename: { "city": "location" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 13,
  modifiedCount: 11,
  upsertedCount: 0
}
```

```
>_MONGOSH
}
> db.students.updateOne(
  { rollNo: 101 },
  { $set: { attendance: 90 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.updateOne(
  { rollNo: 101 },
  { $unset: { attendance: "" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.deleteOne({ rollNo: 103 })
< {
```

```
>_MONGOSH
  acknowledged: true,
  deletedCount: 1
}
> db.students.deleteMany({ marks: { $lt: 40 } })
< {
  acknowledged: true,
  deletedCount: 0
}
> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 12
}
> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 0
}
> db.students.find().sort({ marks: -1 })
<
> db.students.insertMany([
  { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
  { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
  { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
  { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
  { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
  { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
```

```
>_MONGOSH
{ rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
{ rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
{ rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
{ rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a94efacdce903635e4dd609'),
    '1': ObjectId('6a94efacdce903635e4dd60a'),
    '2': ObjectId('6a94efacdce903635e4dd60b'),
    '3': ObjectId('6a94efacdce903635e4dd60c'),
    '4': ObjectId('6a94efacdce903635e4dd60d'),
    '5': ObjectId('6a94efacdce903635e4dd60e'),
    '6': ObjectId('6a94efacdce903635e4dd60f'),
    '7': ObjectId('6a94efacdce903635e4dd610'),
    '8': ObjectId('6a94efacdce903635e4dd611'),
    '9': ObjectId('6a94efacdce903635e4dd612')
  }
}
> db.students.find().sort({ marks: 1 })
< {
  _id: ObjectId('6a94efacdce903635e4dd60f'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
```

```
>_MONGOSH
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60b'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60e'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd609'),
    rollNo: 101,
    name: 'John',
```

>_MONGOSH

```
    age: 20,  
    department: 'CSE',  
    marks: 85,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd611'),  
    rollNo: 109,  
    name: 'Arun',  
    age: 22,  
    department: 'AI&DS',  
    marks: 87,  
    city: 'Kurnool'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd60c'),  
    rollNo: 104,  
    name: 'Priya',  
    age: 22,  
    department: 'IT',  
    marks: 88,  
    city: 'Warangal'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd60a'),  
    rollNo: 102,
```

>_MONGOSH

```
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94efacdce903635e4dd610'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd612'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60d'),
```

```
>_MONGOSH
```

```
    rollNo: 105,  
    name: 'Kiran',  
    age: 21,  
    department: 'CSE',  
    marks: 95,  
    city: 'Hyderabad'  
  }  
> db.students.find().sort({ department: 1, marks: -1 })  
< {  
  _id: ObjectId('6a94efacdce903635e4dd612'),  
  rollNo: 110,  
  name: 'Meena',  
  age: 20,  
  department: 'AI&DS',  
  marks: 93,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd611'),  
  rollNo: 109,  
  name: 'Arun',  
  age: 22,  
  department: 'AI&DS',  
  marks: 87,  
  city: 'Kurnool'  
}  
r
```

>_MONGOSH

```
{
  _id: ObjectId('6a94efacdce903635e4dd60d'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd610'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60a'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
```

>_MONGOSH

```
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd609'),  
  rollNo: 101,  
  name: 'John',  
  age: 20,  
  department: 'CSE',  
  marks: 85,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60b'),  
  rollNo: 103,  
  name: 'Rahul',  
  age: 20,  
  department: 'ECE',  
  marks: 75,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60f'),  
  rollNo: 107,  
  name: 'Ravi',  
  age: 23,  
  department: 'ECE',  
  marks: 68,
```

>_MONGOSH

```
    city: 'Nellore'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60c'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60e'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  }
> db.students.find().limit(5)
< {
  _id: ObjectId('6a94efacdce903635e4dd609'),
  rollNo: 101,
  name: 'John',
  age: 20,
```

>_MONGOSH

```
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60a'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60b'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60c'),
    rollNo: 104,
    name: 'Priya',
```

```
>_MONGOSH
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60d'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
> db.students.find().sort({ marks: -1 }).limit(3)
< {
  _id: ObjectId('6a94efacdce903635e4dd60d'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd612'),
  rollNo: 110
```

>_MONGOSH

```
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd610'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  }
> db.students.find().skip(5)
< {
  _id: ObjectId('6a94efacdce903635e4dd60e'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
```

>_MONGOSH

```
{
  _id: ObjectId('6a94efacdce903635e4dd60f'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a94efacdce903635e4dd610'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd611'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
```

>_MONGOSH

```
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd612'),  
  rollNo: 110,  
  name: 'Meena',  
  age: 20,  
  department: 'AI&DS',  
  marks: 93,  
  city: 'Hyderabad'  
}  
> db.students.find().skip(2).limit(3)  
< {  
  _id: ObjectId('6a94efacdce903635e4dd60b'),  
  rollNo: 103,  
  name: 'Rahul',  
  age: 20,  
  department: 'ECE',  
  marks: 75,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60c'),  
  rollNo: 104,  
  name: 'Priya',  
  age: 22,  
  department: 'IT',
```

```
>_MONGOSH
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60d'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
> db.students.find(
  {},
  { _id: 0, name: 1, marks: 1 }
)
< {
  name: 'John',
  marks: 85
}
{
  name: 'Alice',
  marks: 90
}
{
  name: 'Rahul',
  marks: 75
}
```

>_MONGOSH

```
}  
{  
  name: 'Priya',  
  marks: 88  
}  
{  
  name: 'Kiran',  
  marks: 95  
}  
{  
  name: 'Anu',  
  marks: 82  
}  
{  
  name: 'Ravi',  
  marks: 68  
}  
{  
  name: 'Sita',  
  marks: 91  
}  
{  
  name: 'Arun',  
  marks: 87  
}  
{
```

```
>_MONGOSH
{
  name: 'Meena',
  marks: 93
}
> db.students.find(
  {},
  { _id: 0, name: 1, department: 1, marks: 1 }
)
< {
  name: 'John',
  department: 'CSE',
  marks: 85
}
{
  name: 'Alice',
  department: 'CSE',
  marks: 90
}
{
  name: 'Rahul',
  department: 'ECE',
  marks: 75
}
{
  name: 'Priya',
  department: 'IT',
  marks: 88
}
```

>_MONGOSH

```
}  
{  
  name: 'Kiran',  
  department: 'CSE',  
  marks: 95  
}  
{  
  name: 'Anu',  
  department: 'IT',  
  marks: 82  
}  
{  
  name: 'Ravi',  
  department: 'ECE',  
  marks: 68  
}  
{  
  name: 'Sita',  
  department: 'CSE',  
  marks: 91  
}  
{  
  name: 'Arun',  
  department: 'AI&DS',  
  marks: 87  
}
```

```
>_MONGOSH
```

```
{
  name: 'Meena',
  department: 'AI&DS',
  marks: 93
}
> db.students.find(
  {},
  { age: 0 }
)
< {
  _id: ObjectId('6a94efacdce903635e4dd609'),
  rollNo: 101,
  name: 'John',
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60a'),
  rollNo: 102,
  name: 'Alice',
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
```

>_MONGOSH

```
{
  _id: ObjectId('6a94efacdce903635e4dd60b'),
  rollNo: 103,
  name: 'Rahul',
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60c'),
  rollNo: 104,
  name: 'Priya',
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60d'),
  rollNo: 105,
  name: 'Kiran',
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60e'),
```

>_MONGOSH

```
  rollNo: 106,  
  name: 'Anu',  
  department: 'IT',  
  marks: 82,  
  city: 'Guntur'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60f'),  
  rollNo: 107,  
  name: 'Ravi',  
  department: 'ECE',  
  marks: 68,  
  city: 'Nellore'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd610'),  
  rollNo: 108,  
  name: 'Sita',  
  department: 'CSE',  
  marks: 91,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd611'),  
  rollNo: 109,  
  name: 'Arun',  
  department: 'AT&DS',
```

>_MONGOSH

```
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd612'),
    rollNo: 110,
    name: 'Meena',
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  }
> db.students.distinct("department")
< [ 'AI&DS', 'CSE', 'ECE', 'IT' ]
> db.students.distinct("city")
< [
  'Guntur',
  'Hyderabad',
  'Kurnool',
  'Nellore',
  'Vijayawada',
  'Warangal'
]
> db.students.distinct(
  "department",
  { marks: { $gte: 80 } }
)
```

```
>_MONGOSH
{ marks: { type: 'number' } }
)
< [ 'AI&DS', 'CSE', 'IT' ]
> db.students.createIndex({ rollNo: 1 })
< rollNo_1
> db.students.createIndex({ name: 1 })
< name_1
> db.students.createIndex({ department: 1, marks: -1 })
< department_1_marks_-1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { rollNo: 1 }, name: 'rollNo_1' },
  { v: 2, key: { name: 1 }, name: 'name_1' },
  {
    v: 2,
    key: { department: 1, marks: -1 },
    name: 'department_1_marks_-1'
  }
]
> db.students.dropIndex("name_1")
< { nIndexesWas: 4, ok: 1 }
> db.students.dropIndexes()
< {
  nIndexesWas: 3,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}
```

```
>_MONGOSH
}
> db.students.aggregate([
  { $match: {} }
])
< {
  _id: ObjectId('6a94efacdce903635e4dd609'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60a'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60b'),
  rollNo: 103,
  name: 'Rahul',
```

>_MONGOSH

```
  age: 20,  
  department: 'ECE',  
  marks: 75,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60c'),  
  rollNo: 104,  
  name: 'Priya',  
  age: 22,  
  department: 'IT',  
  marks: 88,  
  city: 'Warangal'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60d'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60e'),  
  rollNo: 106,
```

>_MONGOSH

```
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60f'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a94efacdce903635e4dd610'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd611'),
```

```
>_MONGOSH
```

```
  rollNo: 109,  
  name: 'Arun',  
  age: 22,  
  department: 'AI&DS',  
  marks: 87,  
  city: 'Kurnool'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd612'),  
  rollNo: 110,  
  name: 'Meena',  
  age: 20,  
  department: 'AI&DS',  
  marks: 93,  
  city: 'Hyderabad'  
}  
> db.students.aggregate([  
  { $match: { marks: { $gt: 80 } } }  
])  
< {  
  _id: ObjectId('6a94efacdce903635e4dd609'),  
  rollNo: 101,  
  name: 'John',  
  age: 20,  
  department: 'CSE',  
  marks: 85,
```

>_MONGOSH

```
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60a'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60c'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  }
  {
    _id: ObjectId('6a94efacdce903635e4dd60d'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
```

>_MONGOSH

```
marks: 95,  
city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd60e'),  
  rollNo: 106,  
  name: 'Anu',  
  age: 20,  
  department: 'IT',  
  marks: 82,  
  city: 'Guntur'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd610'),  
  rollNo: 108,  
  name: 'Sita',  
  age: 21,  
  department: 'CSE',  
  marks: 91,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a94efacdce903635e4dd611'),  
  rollNo: 109,  
  name: 'Arun',  
  age: 22,
```

```
>_MONGOSH

  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a94efacdce903635e4dd612'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      totalStudents: { $sum: 1 }
    }
  }
])
< {
  _id: 'AI&DS',
  totalStudents: 2
}
{
  _id: 'IT',
```

>_MONGOSH

```
{
  _id: 'IT',
  totalStudents: 2
}
{
  _id: 'CSE',
  totalStudents: 4
}
{
  _id: 'ECE',
  totalStudents: 2
}
> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" }
    }
  }
])
< {
  _id: 'AI&DS',
  averageMarks: 90
}
{
  _id: 'IT',
  averageMarks: 85
}
```

>_MONGOSH

```
  _id: 'IT',
  averageMarks: 85
}
{
  _id: 'CSE',
  averageMarks: 90.25
}
{
  _id: 'ECE',
  averageMarks: 71.5
}
> db.students.aggregate([
  {
    $group: {
      _id: null,
      highestMarks: { $max: "$marks" }
    }
  }
])
< {
  _id: null,
  highestMarks: 95
}
> db.students.aggregate([
  {
    $group: {
```

>_MONGOSH

```
    _id: null,
    lowestMarks: { $min: "$marks" }
  }
}
])
< {
  _id: null,
  lowestMarks: 68
}
> db.students.aggregate([
  {
    $group: {
      _id: null,
      totalMarks: { $sum: "$marks" }
    }
  }
])
< {
  _id: null,
  totalMarks: 854
}
> db.students.aggregate([
  { $match: { department: "CSE" } },
  {
    $group: {
      _id: "$department",
```

>_MONGOSH

```
    averageMarks: { $avg: "$marks" },
    totalStudents: { $sum: 1 }
  }
},
{ $sort: { averageMarks: -1 } }
])
< {
  _id: 'CSE',
  averageMarks: 90.25,
  totalStudents: 4
}
> db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 },
  {
    $project: {
      _id: 0,
      rollNo: 1,
      name: 1,
      marks: 1
    }
  }
])
< {
  rollNo: 105,
  name: 'Kiran',
```

```
>_MONGOSH
```

```
    marks: 95
  }
  {
    rollNo: 110,
    name: 'Meena',
    marks: 93
  }
  {
    rollNo: 108,
    name: 'Sita',
    marks: 91
  }
}
```

```
> show dbs
```

```
< admin          40.00 KiB
  bookflow_db    100.00 KiB
  config         100.00 KiB
  local          72.00 KiB
  studentDB      48.00 KiB
```

```
> show collections
```

```
< students
```

```
> db
```

```
< studentDB
```

```
> db.students.find().pretty()
```

```
< {
  _id: ObjectId('6a94efacdce903635e4dd609'),
  rollNo: 101,
```

>_MONGOSH

```
    age: 20,  
    department: 'CSE',  
    marks: 85,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd60a'),  
    rollNo: 102,  
    name: 'Alice',  
    age: 21,  
    department: 'CSE',  
    marks: 90,  
    city: 'Vijayawada'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd60b'),  
    rollNo: 103,  
    name: 'Rahul',  
    age: 20,  
    department: 'ECE',  
    marks: 75,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd60c'),  
    rollNo: 104,
```

>_MONGOSH

```
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60d'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60e'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a94efacdce903635e4dd60f'),
```

>_MONGOSH

```
    rollNo: 107,  
    name: 'Ravi',  
    age: 23,  
    department: 'ECE',  
    marks: 68,  
    city: 'Nellore'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd610'),  
    rollNo: 108,  
    name: 'Sita',  
    age: 21,  
    department: 'CSE',  
    marks: 91,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a94efacdce903635e4dd611'),  
    rollNo: 109,  
    name: 'Arun',  
    age: 22,  
    department: 'AI&DS',  
    marks: 87,  
    city: 'Kurnool'  
  }  
  {
```

```
>_MONGOSH
{
  _id: ObjectId('6a94efacdce903635e4dd612'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
> db.students.countDocuments()
< 10
> db.students.distinct("department")
< [ 'AI&DS', 'CSE', 'ECE', 'IT' ]
> db.students.getIndexes()
< [ { v: 2, key: { _id: 1 }, name: '_id_' } ]
> use studentDB
< already on db studentDB
> db.students.insertOne({
  rollNo: 201,
  name: "Test Student",
  age: 20,
  department: "CSE",
  marks: 85
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a94f19ddce903635e4dd613')
```

```
>_MONGOSH
```

```
    }  
> db.students.find({ rollNo: 201 })  
< {  
  _id: ObjectId('6a94f19ddce903635e4dd613'),  
  rollNo: 201,  
  name: 'Test Student',  
  age: 20,  
  department: 'CSE',  
  marks: 85  
}  
> db.students.updateOne(  
  { rollNo: 201 },  
  { $set: { marks: 95 } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
> db.students.deleteOne({ rollNo: 201 })  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

```
> db.students.deleteOne({ rollNo: 201 })  
< {  
  acknowledged: true,  
  deletedCount: 1  
}  
>  
studentDB> |
```